Rank-27 Post-Vivify TransitiveReduce Scheduling Repair

Tactix engineering handoff

21 August 2026

Objective and outcome

The requested work was to finish the Vivify parity investigation—including candidate scheduling, literal order, and interleaving with surrounding proof rules—then report the verified result. The Vivify repair itself remains closed: candidate scope, cumulative tier budgets, jumped-reason deduction, terminal effects, and productivity match the retained Kissat evidence. The first portable divergence after those calls was the TransitiveReduce pass between them.

Tactix processed V4 sources in ascending literal order. Kissat activity-sorts them, and its literal source contains an unusual but observable detail: the stack holds variable indices, while the comparator applies `IDX` again before reading EVSIDS/VMTF activity, sorts increasing, then pops from the back. Tactix now poses the equivalent rule-local order for its front-consumed array. No input name, hash, rank, literal, deadline, or tuned constant enters the solver. Early BFS exit was tested separately and left unchanged because it was inert at the localized boundary.

Release passes 72/72. The required matched 28B+5C guard is 66/66 valid, with all SAT models and UNSAT proofs checked and no certificate residue. The new rank-27 witness still returns UNKNOWN at 1,200.004 seconds, so this is a source/scheduling repair rather than a Tier-D closure.

Frozen identities and experiment discipline

The immutable expected-UNSAT rank-27 input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, compressed SHA-256 `5435b1b2...bb4e9`. The accepted binary is `b6c991e0...fb6f`, source `7b527fbc...b4b`. The candidate is `4caa9276...02a0`, built from base `62f3bca8...0b97` plus source diff `110eafdf...ac85`.

The branch and `origin/main` were synchronized before work. Surviving worktrees and unrelated SLURM jobs were inspected and left untouched. The frozen rank-21–40 manifest, six inherited closures, and prior reports were read first. No frozen reference solver was rerun: retained Kissat traces and Tactix-only source-pinned diagnostics were sufficient. Diagnostics used two CPUs, 16 GiB, and 300 seconds; the guard and witness used the pre-Phase-4.5 16 GiB/1,200-second contract.

Localization from the matched Vivify state

The repaired trace reaches the same conflict-3069 Probe trigger and restores both Vivify long-clause populations and aggregate terminal effects. The first later decision mismatch was traced backward from decision level 35. EVSIDS scores matched, but heap topology did not: the Probe pipeline removed 369 Kissat variables versus 33 Tactix variables. Slot snapshots made the first formula owner the TransitiveReduce call between the Vivify calls.

The retained source comparison separated three mechanics: mode-dependent activity order versus ascending literals, early destination exit versus a complete BFS, and Kissat's second `IDX` conversion on a scheduled variable index. Two bounded Tactix-only experiments isolated them.

Job `1637470` tested native order, early exit alone, an ordinary own-variable heuristic order, and their combination. Early exit was bit-identical. The ordinary heuristic changed work but deleted no binaries at the boundary, so that candidate was reverted.

Job `1637474` then posed the literal reference comparator, both with and without early exit. The two arms were identical. The first V4 call changed from zero deletions to 24; the retained Kissat trace deletes 21. Its ending binary counts are 751,109 Tactix and 751,102 Kissat, reducing a ten-binary pre-call excess to seven. After the second Vivify they are 751,122 and 751,112. Later Sweep/Substitute transitions make the counts close through different equivalence and retained-root representations. That is the next open boundary; the evidence does not reopen Vivify deduction or V4 source order.

Implementation decision

The production change is deliberately narrow.

- Only TransitiveReduce interprets its heuristic arm through the reference `v » 1` activity proxy and back-to-front tie semantics.
- Because Tactix consumes from the front, it uses decreasing activity and a decreasing-variable tie, preserving positive-then-negative sign order.
- The default V4 slot selects this order. Other rules and candidate-order arms retain their prior meaning.
- Early exit remains off because its one-at-a-time arm changed no formula state at the witness boundary.

The focused regression independently computes the best reference proxy and requires that candidate first. The slot test requires V4 to ship with the reference activity arm. Release's 72 tests include the TransitiveReduce brute-force equivalence oracle and proof checks. No unit test or fuzzing exposed a crash or soundness defect, so the plan's mandatory Claude repair-delegation trigger did not fire.

Correctness and performance guard

Matched campaign `benchmarks/runs/20260821-1830-BC-transitive-reference-order/` ran the accepted pin and candidate on all 28 Tier-B fixtures plus the inherited fixed five mechanism-relevant Tier-C fixtures.

The first submission, job `1637476`, was invalid harness setup: `matrix.tsv` had not been copied. All tasks failed before CNF decompression or solver invocation, and produced no certificate residue. The unchanged frozen matrix was restored; corrected job `1637542` completed.

Arm	valid	SAT / UNSAT	aggregate	peak RSS
Accepted pin	33/33	17 / 16	2202.399 s	3.234 GiB
Candidate	33/33	17 / 16	2773.863 s	3.360 GiB

All 34 SAT models and 32 UNSAT proofs verify. There are no invalid, malformed, missing, cleanup-bad, or residual-certificate tasks. The ratio is 1.259473, so the earlier `mp1-Nb5T15`/`sted5` performance coupling remains open. This faithful repair is not promoted as a new accepted performance pin.

Rank-27 witness

Only after the full guard passed was immutable witness job `1637608` submitted. It returned UNKNOWN at 1,200.004 seconds. The runner recorded a 147,849,216-byte partial proof, SHA-256 `38b4c082...85033`, then deleted it and all derived scratch. Certificate residue is zero. The previous Vivify-only witness generated a 252,706,816-byte partial proof at the same deadline; the smaller artifact is descriptive trajectory evidence, not a solved result or timing conclusion.

Limitations and next action

Rank 27 remains open and is not added to the nine closed ranks. The next UNSAT-first localization begins at the residual post-Vivify Sweep/equivalence and retained-root representation split. It must not repeat the completed Vivify candidate/deduction work or the closed TransitiveReduce ordering attribution. In parallel sequence, the 1.259x fixed-C/Tier-B coupling remains subject to mechanism localization. Any further source repair again owes the complete 28B+fixed-C guard before another Tier-D witness.

The durable technical report is in `benchmarks/reports/`, file `2026-08-21-tierd-rank27-transitive-reference-order.md`. The current resume surface is `PROGRESS.md`; durable mechanism evidence is in `docs/implementation-notes.md` and `docs/scheduling-parity-ledger.md`.